

## Growing Instead of Gathering

The earliest adoption of new foods took place as human societies spread out of Africa into new environments inhabited by unfamiliar plants and animals. Imagine the first humans in Australia, roughly 60,000 years ago, encountering giant kangaroos (<https://en.wikipedia.org/wiki/Procoptodon>) up to 2.7 meters (nearly 9 feet) in height, sheep-sized echidnas (<https://en.wikipedia.org/wiki/Murrayglossus>), and wombats the size of a hippopotamus (<https://en.wikipedia.org/wiki/Diprotodon>)! Not to mention plants unique to Australia, many of which are poisonous or harmful, such as the "stinging tree" ([https://en.wikipedia.org/wiki/Dendrocnide\\_moroides](https://en.wikipedia.org/wiki/Dendrocnide_moroides)). Exploring this landscape took time and careful study of plants to understand which were valuable and which were harmful. But once a society has built this body of knowledge, passing it on from generation to generation, then hunters know well where to find prey, and useful plants can be gathered in all seasons.

Once this body of knowledge exists, why make a shift to raising food? One plausible argument is that once a society commits to agriculture, the resulting food surplus means that more children can be born and are also needed to help with farming and herding; this leads to population-growth rates much higher than those of societies committed to hunting and gathering. As agricultural populations grew, they expanded onto new land, pushing aside existing populations who subsisted on hunting and gathering. But it is also possible that hunters and gatherers learned how to farm and adopted some aspects of agriculture into their societies. So, how did agriculture spread across large areas of the world? As Gremillion describes, there is evidence that both hypotheses are correct. When compared with genetic analysis of archaeological human remains that predate agriculture, genetic analysis of modern people from across Europe—one of the earliest areas into which agriculture spread—indicates that modern populations include genetic ancestry from both groups of people: pre-agricultural populations of Europe and agricultural colonists from Anatolia (what is today Turkey).

The wholesale replacement of hunting and gathering with agriculture, however, did not occur everywhere. In some cases, societies only adopted specific cultivated plants or animals, ignoring others. One example is the societies of the Pacific Northwest (today coastal Oregon, Washington, and British Columbia), who lived in dense populations along coasts and rivers of this region, in which fish—especially salmon—played a large role in subsistence. But despite the importance of fish and gathered plants to their diet, they cultivated several native root crops—some of which can be considered domesticated—and grew tobacco, an early cultivated crop that spread across North America; concurrently, they ignored maize, squash, and other domesticates common to diets

across North America prior to the arrival of Europeans. It has been argued that this was because local gathered resources and root crops were already sufficient and well adapted to local conditions, making maize and other crops unnecessary, but the answer may be more culturally specific. Tobacco and local root crops had specific roles in feasting ceremonies that were central to social life across the region. Maybe maize and other foreign crops were ignored because they did not fit into this social fabric and so had no value.